PROJECT DOCUMENTATION

Student Enrollment & Course Management System

(SECMS)

Salesforce-Based Academic Enrollment Management System

PRESENTED BY

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Abstract

The Student Enrollment & Course Management System (SECMS) is a Salesforce-based application for organizing student, course, instructor, and enrollment information in one academic workspace. The project models these records as connected Salesforce objects and uses validation rules, approval processing, and Flow automation to support the enrollment lifecycle. Reports and dashboards provide views of student status, course enrollment, pending approvals, and fee information. This revised documentation follows the original project topics and incorporates the supplied Salesforce screenshots as implementation evidence. The screenshots illustrate configured objects and fields, automation setup, approval configuration, reports, and dashboards. The implementation is presented as a Salesforce Developer Edition project; observed sample data and visible configuration are described conservatively.

Keywords: Salesforce, student enrollment, custom objects, Flow, approval process, reports, dashboards.

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1. Introduction

1.1 Project Overview

SECMS centralizes the records and routine work involved in academic enrollment. Its core data entities are Student, Course, Instructor, and Enrollment. Enrollment records connect a student to a selected course and can also identify the assigned instructor. Salesforce Lightning provides the application interface, while platform configuration supports data entry, review, automation, and reporting.

1.2 User Story

As an enrollment officer or academic administrator, I want student, course, instructor, and enrollment details in one system so that requests can be processed consistently and their status can be reviewed without searching through separate records.

1.3 Purpose and Objectives

- Maintain connected academic records in a common Salesforce application.
- Use validation and approval steps to support consistent enrollment processing.
- Automate repeatable actions such as setting dates, notifying users, and creating follow-up tasks.
- Use reports and dashboards to monitor student status, enrollments, approvals, and fees.

1.4 Scope

The project covers foundational record management, enrollment status handling, selected Flow automation, approval configuration, reports, dashboards, and role-based access planning. A payment gateway, student self-service portal, production identity integration, and institution-wide deployment are considered future enhancements.

2. Ideation Phase

2.1 Problem Statement

When student, course, and registration details are kept in separate files or email threads, staff can encounter duplicate data entry, incomplete records, delayed approvals, and uncertainty about current enrollment status. A centralized system can connect the main academic records and provide a clearer path from request to review and reporting.

2.2 Empathy Map: Enrollment Officer

Area	Observed need
Think and feel	Wants accurate records and timely decisions; is concerned about duplicate entries and incomplete requests.
Hear	Receives questions and follow-ups from students, instructors, and management.
See	Works with records, status changes, approval queues, and updates from multiple sources.
Say and do	Checks record details, follows up on requests, and updates enrollment information.
Pain points	Repeated work, delayed review, and limited visibility into pending items.
Desired gains	Connected records, dependable process steps, and useful reporting.

2.3 Ideas Considered

- Spreadsheet-based tracking with manual updates.
- A standalone enrollment website.
- Integration with a larger campus ERP system.
- A Salesforce application using objects, automation, access controls, reports, and dashboards.

The Salesforce approach was selected for this project because it allows data structures, process automation, approvals, and analytics to be configured on one cloud platform. The selection is suitable for a prototype and can be extended as institutional requirements become clearer.

3. Requirement Analysis

3.1 Customer Journey

Stage	System activity
Registration	Capture student details and create a Student record.
Course selection	Maintain Course information and identify the intended course.
Enrollment request	Create an Enrollment record and relate it to the student and course.
Fee handling	Record the fee amount and whether fees have been paid.
Approval	Route eligible requested enrollments for review.
Instructor assignment	Associate an instructor with the enrollment where configured.
Monitoring	Review reports and dashboards for operational status.

3.2 Functional Requirements

ID	Requirement	Description
FR-01	Student management	Create and maintain student information.
FR-02	Course management	Create and maintain course details, category, and fee.
FR-03	Instructor management	Maintain instructor identity and expertise information.
FR-04	Enrollment processing	Create enrollment records and manage their status.
FR-05	Validation and approval	Apply business checks and route requested records for review.
FR-06	Automation	Automate configured enrollment actions and follow-up tasks.
FR-07	Reporting	Provide reports and dashboards for operational monitoring.

3.3 Non-Functional Requirements

- Usability: provide clear navigation and record layouts for common tasks.
- Security: restrict access to academic data according to user responsibilities.
- Reliability: ensure configured validation, approvals, and automation behave consistently.
- Performance: keep record retrieval and reporting responsive for routine use.
- Scalability: allow additional users, courses, departments, and capabilities to be introduced.

4. Project Design

4.1 System Architecture

Users work through Salesforce Lightning pages and tabs. Salesforce custom objects store the project data. Validation Rules and Approval Processes apply configured checks and review steps, while Flows automate actions based on record events. Reports query saved records and dashboards summarize selected reports for administrators.

Design flow: User > Salesforce Lightning > Student / Course / Instructor / Enrollment records > Validation, Flow, and approval actions > Reports and dashboards

4.2 Data Model

Object	Purpose	Representative fields
Student	Stores student identity and contact details.	Student Name, Email, Phone, Date of Birth, Address, Status
Course	Stores course information.	Course Name, Category, Course Fee, Description
Instructor	Stores instructor identity and expertise.	Instructor Name, Instructor Code, Expertise, Email
Enrollment	Connects a student to a course and instructor.	Student, Course, Instructor, Enrollment Date, Enrollment Status, Fees Paid, Total Amount, Comments

4.3 Relationships

Enrollment uses lookup relationships to Student, Course, and Instructor. These links allow users to review related information and support reporting across the enrollment process. The supplied field screenshots show lookup fields from Enrollment to Student, Course, and Instructor.

4.4 Proposed Enrollment Lifecycle

- Create or update the student, course, and instructor records as needed.
- Create an Enrollment and link it to the appropriate records.
- Set its status to Requested and allow configured validation and approval criteria to apply.
- Process the request through approval, then update the status and related information according to the configured outcome.
- Use reports and dashboards to monitor records and outstanding work.

5. Project Planning & Scheduling

5.1 Delivery Approach

The work is organized into iterative configuration stages so that the data model can be established first, followed by process rules, automation, security, and reporting. Each stage should have a clear configuration outcome and be reviewed before moving to the next.

Stage	Planned work	Deliverable
1. Foundation	Create the application and core objects.	Navigation and initial data model
2. Data model	Add fields, relationships, layouts, and tabs.	Usable Student, Course, Instructor, Enrollment records
3. Process controls	Configure validation and approval criteria.	Controlled enrollment review path

Stage	Planned work	Deliverable
4. Automation	Build and activate appropriate record-triggered Flows.	Date, notification, assignment, or follow-up actions
5. Visibility	Configure reports, dashboards, and access.	Operational views and role-appropriate access
6. Review	Run functional checks and document observed results.	Evidence-based project report

5.2 Project Roles

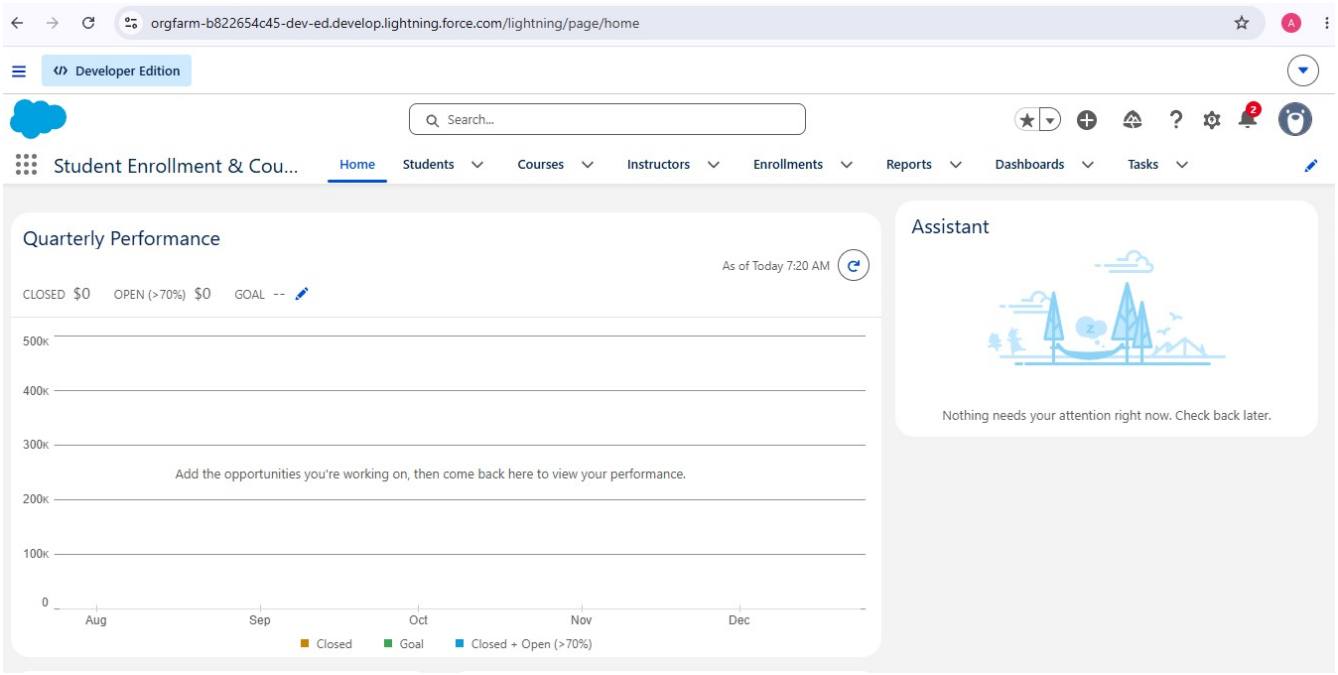
- Project team: analyze requirements, configure Salesforce, capture evidence, and maintain documentation.
- Enrollment officer: create and review student and enrollment records.
- Administrator: manage configuration, reports, access, and approval routing.
- Instructor: view the enrollment information relevant to assigned teaching responsibilities.

6. Project Development & Salesforce Configuration

6.1 Salesforce Application and Objects

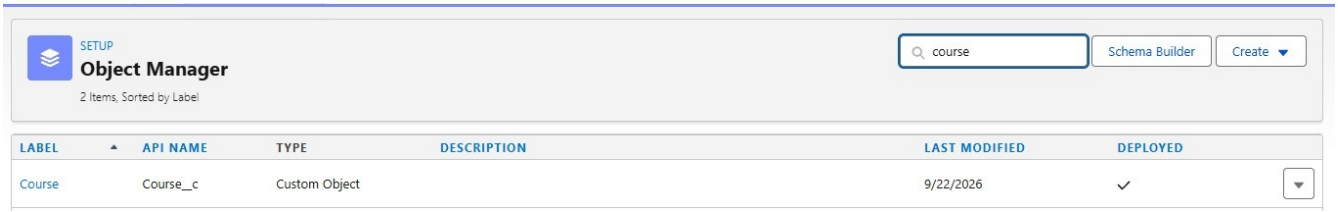
The Salesforce application is presented as "Student Enrollment & Course Management." The project uses the Student, Course, Instructor, and Enrollment custom objects. Object Manager provides the configuration area for object details, fields, relationships, and page behavior.

Evidence 1. Salesforce application home



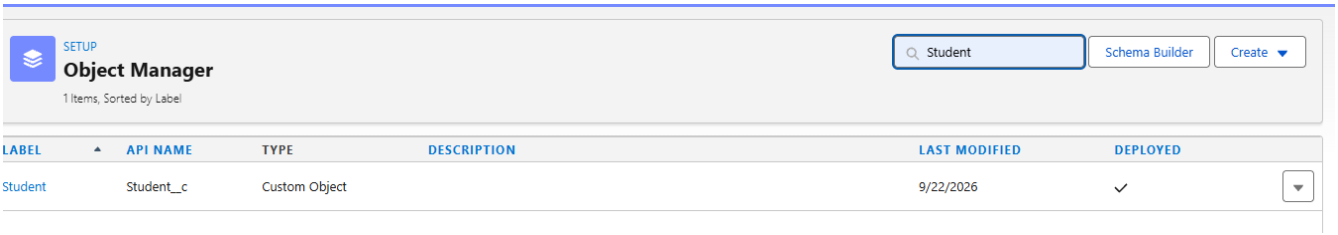
The Lightning application navigation shows the Student, Course, Instructor, Enrollment, Reports, Dashboards, and Tasks areas.

Evidence 2. Course object in Object Manager



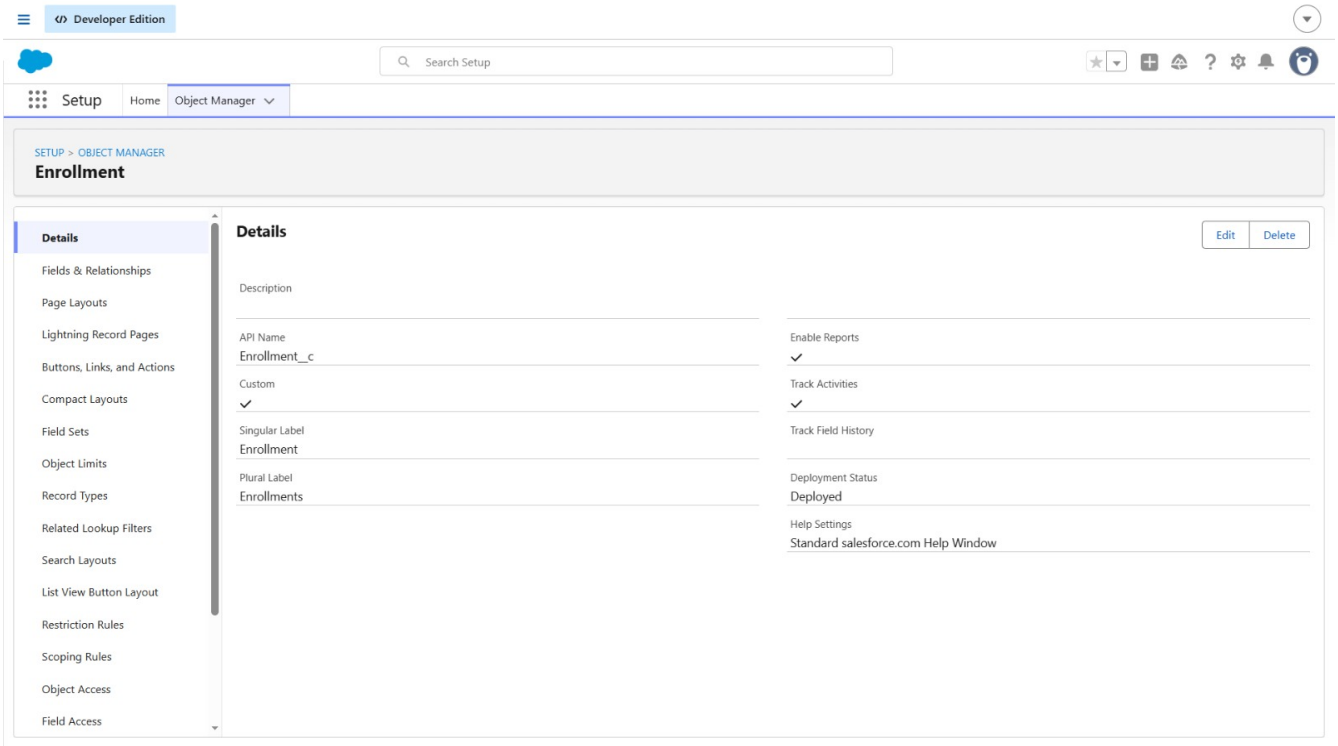
The Course custom object is visible in Object Manager.

Evidence 3. Student object in Object Manager



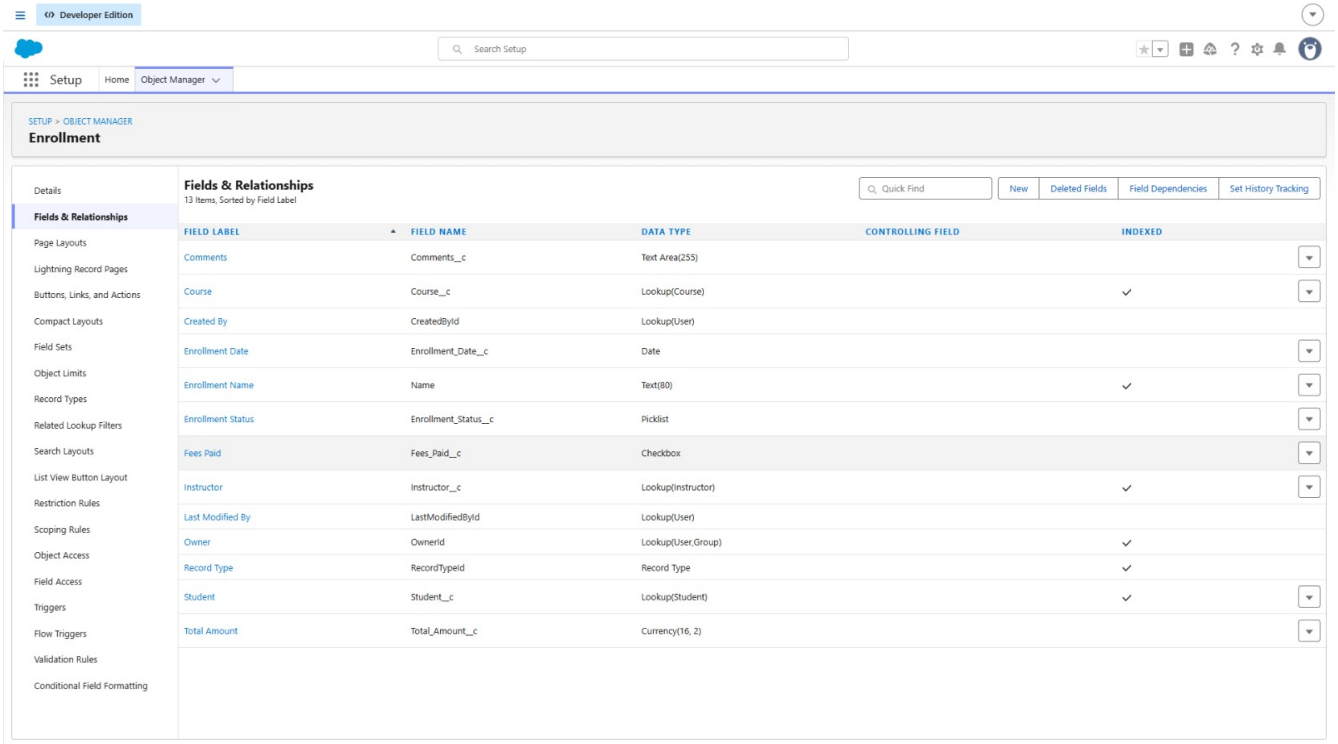
The Student custom object is visible in Object Manager.

Evidence 4. Enrollment object details



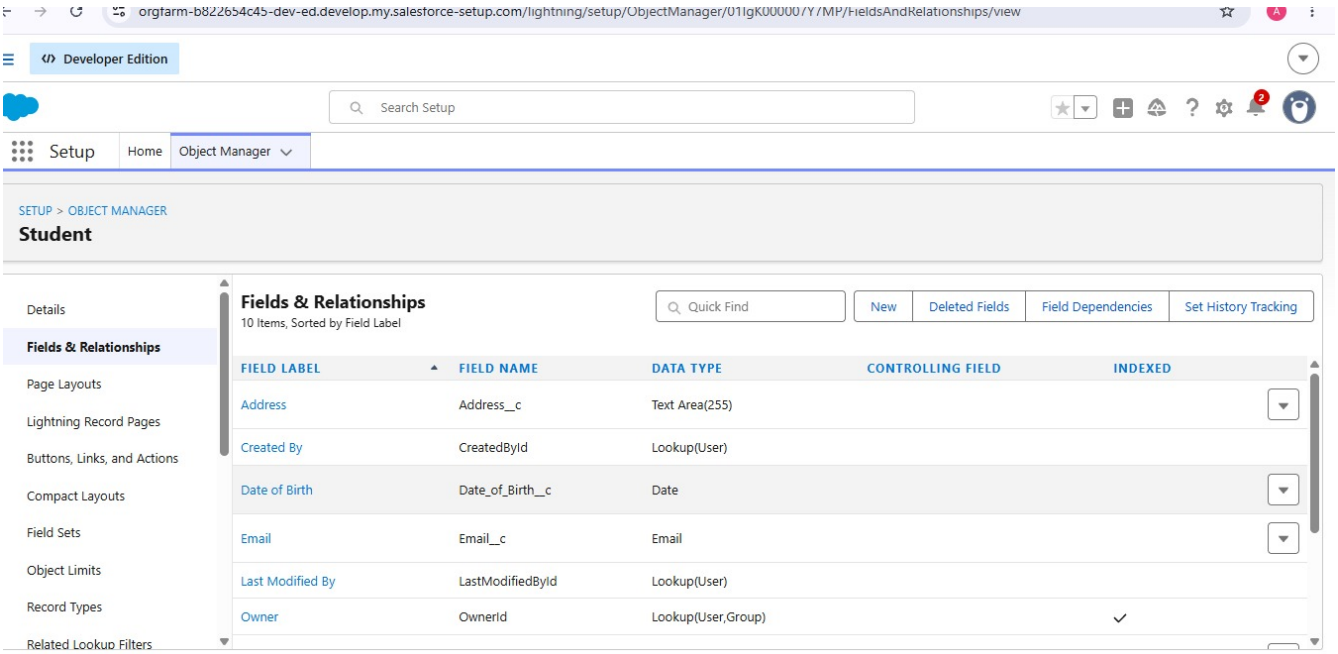
The Enrollment object details page shows its API name, labels, and enabled features.

Evidence 5. Enrollment fields and relationships



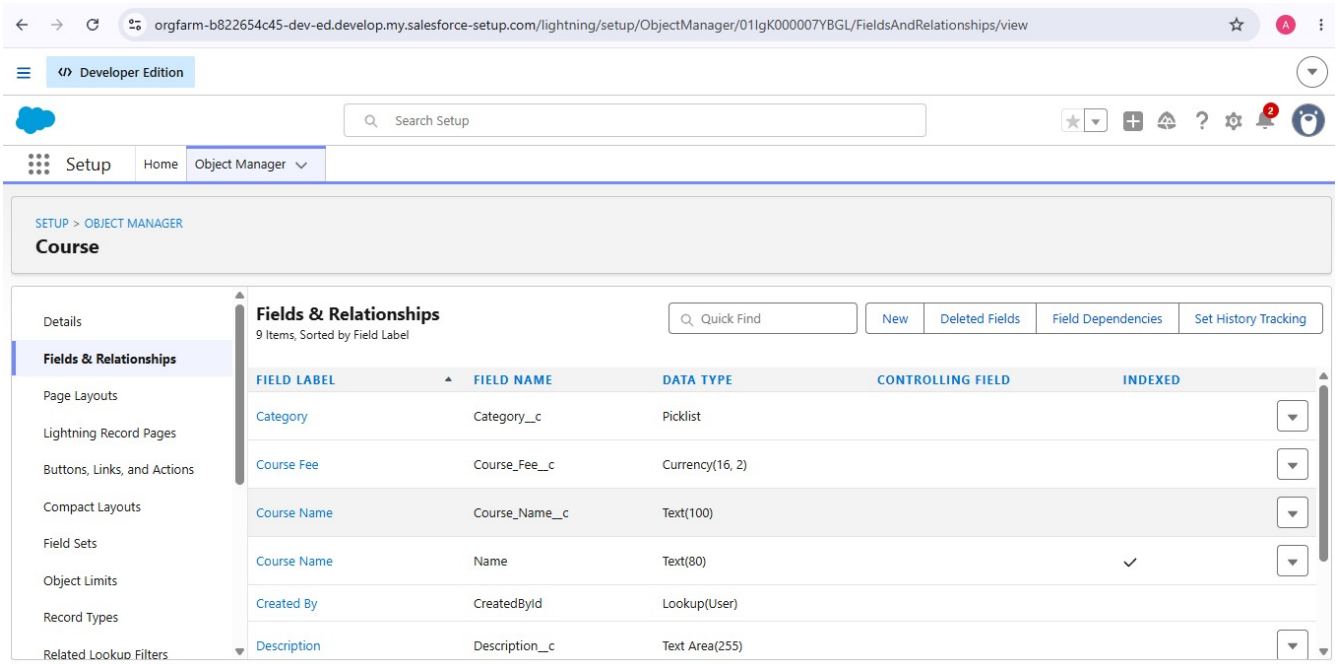
The field list includes student and course lookups, enrollment date and status, instructor, fees paid, and total amount.

Evidence 6. Student fields



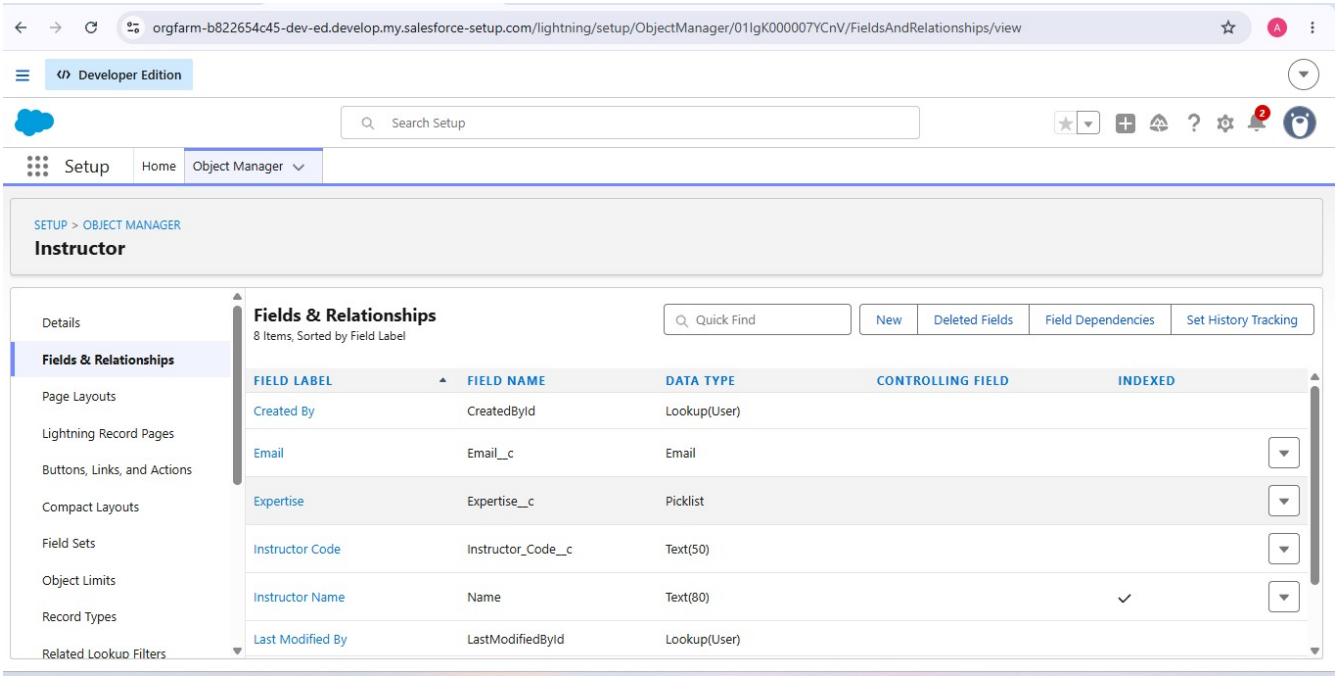
The Student object includes fields such as address, date of birth, and email.

Evidence 7. Course fields



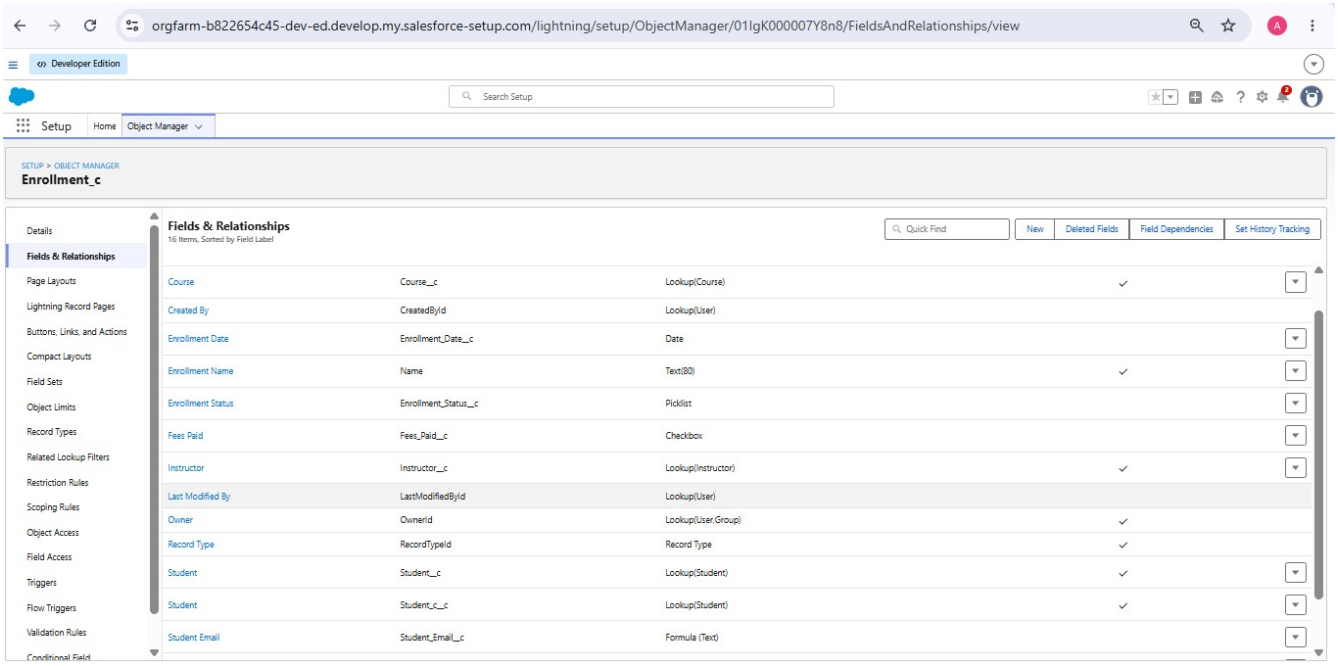
The Course object includes category, fee, name, and description fields.

Evidence 8. Instructor fields



The Instructor object includes email, expertise, instructor code, and instructor name fields.

Evidence 9. Enrollment fields



The Enrollment object field list shows course, date, status, fee, instructor, student, and formula-related fields.

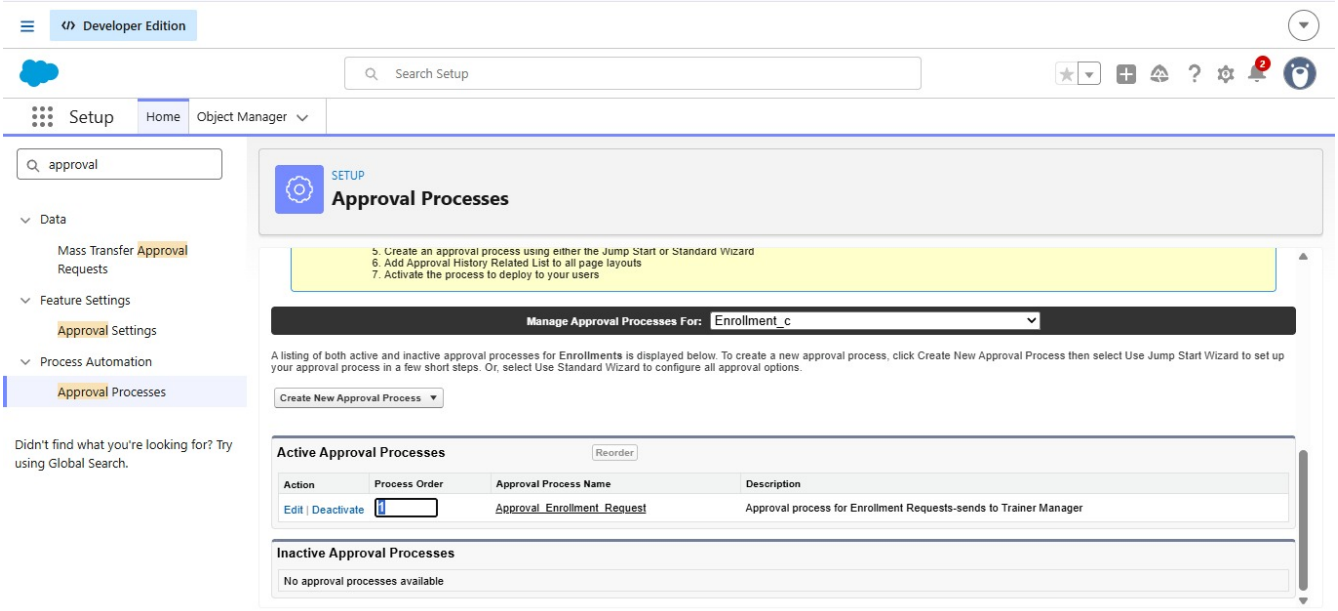
6.2 Data Validation and Record Types

Validation Rules should communicate actionable errors and block records that do not meet the agreed business conditions. The project design includes an institutional email-domain check and a fee-paid check before approval. Record types and layouts can distinguish a first-time enrollment from a re-enrollment, with fields tailored to each process.

6.3 Approval Process

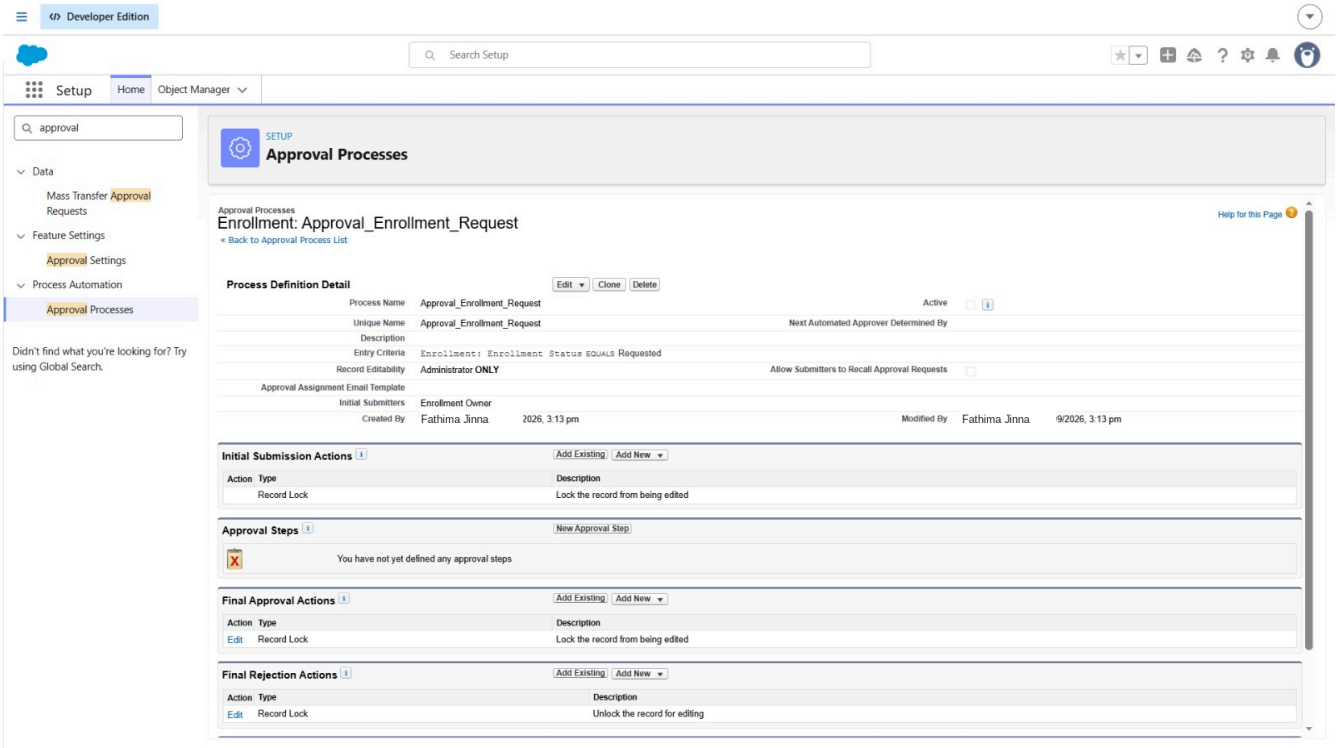
The provided configuration identifies an approval process named Approval_Enrollment_Request for Enrollment records. Its entry criterion is Enrollment Status equals Requested. The process page shows record locking at submission and final record-lock actions. The supplied detail screenshot also shows that no approval step had yet been defined at the time that image was captured; therefore, the documentation treats the routing step and final status updates as intended configuration requiring completion and verification.

Evidence 10. Approval process list



The Enrollment approval process is listed as active, with an order and description for enrollment requests.

Evidence 11. Approval process details

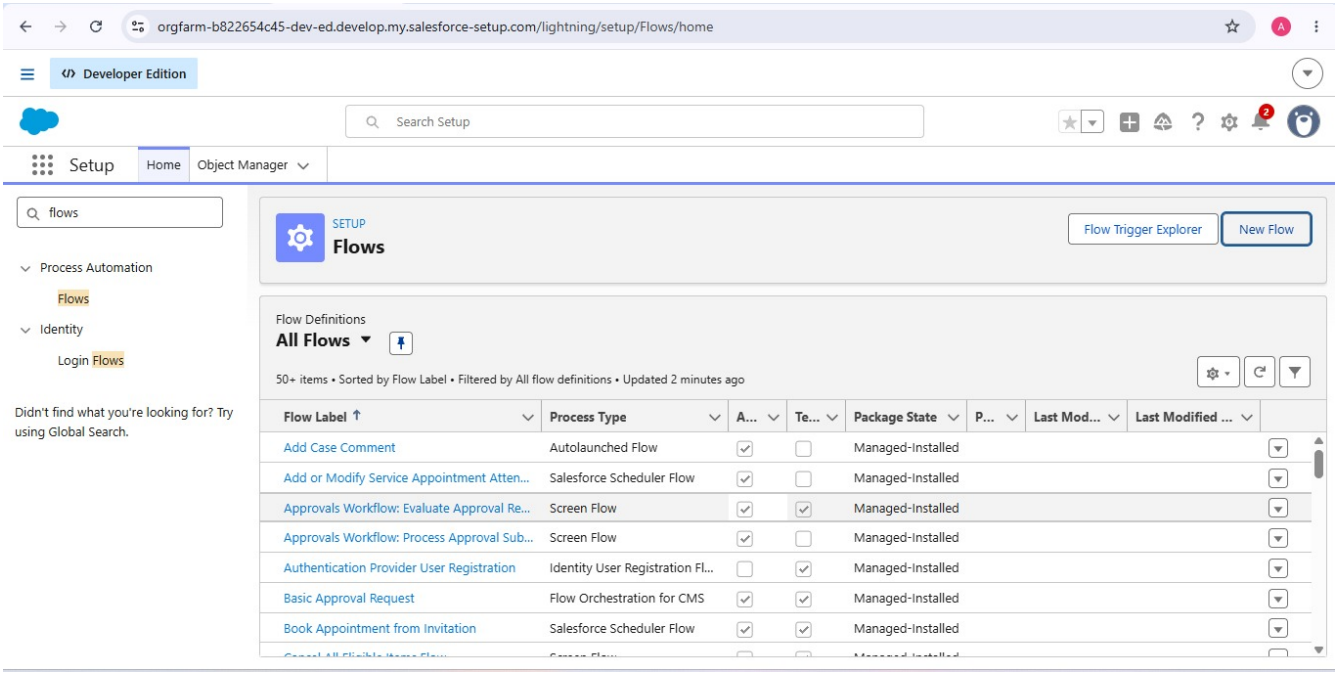


The detail page shows the Requested entry criterion and record-lock actions; it also indicates that an approval step was not yet defined in this captured configuration.

6.4 Flow Automation

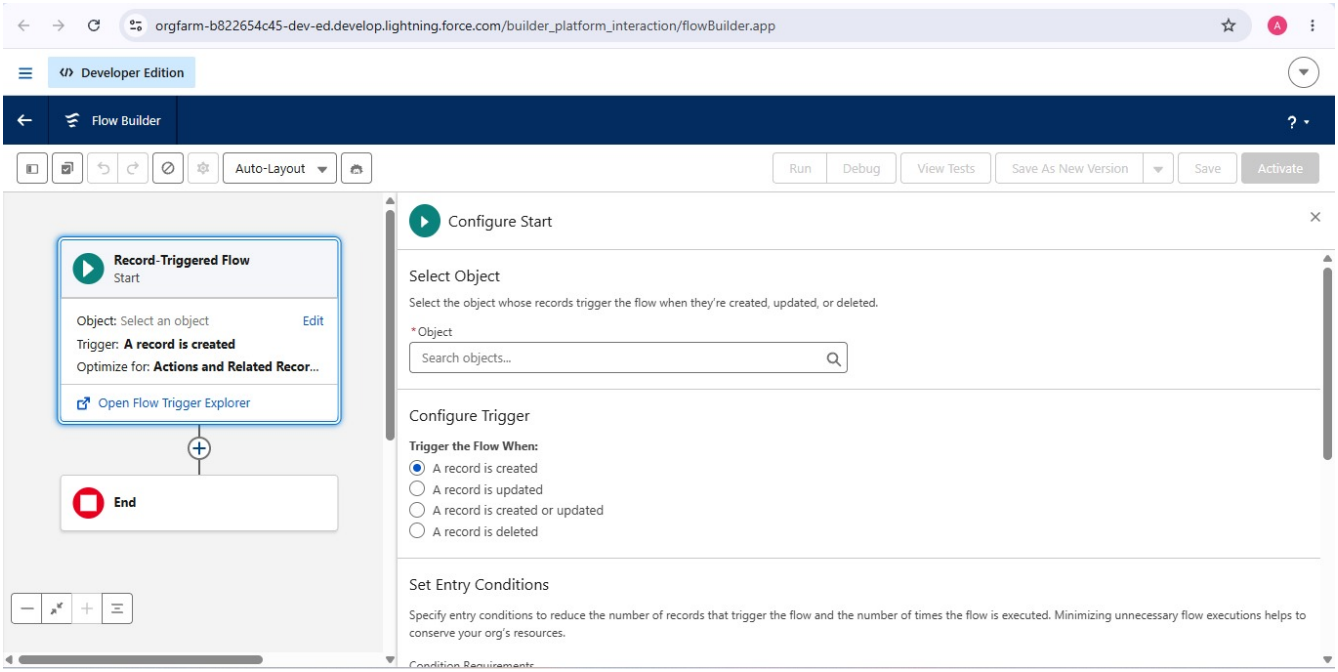
Record-triggered Flows can support repeatable enrollment tasks. The project design considers setting an enrollment date at creation, responding to approved records, assigning an instructor based on course category, and creating an administrator follow-up task for requested enrollments. Each Flow should be tested with representative records before activation.

Evidence 12. Flow list



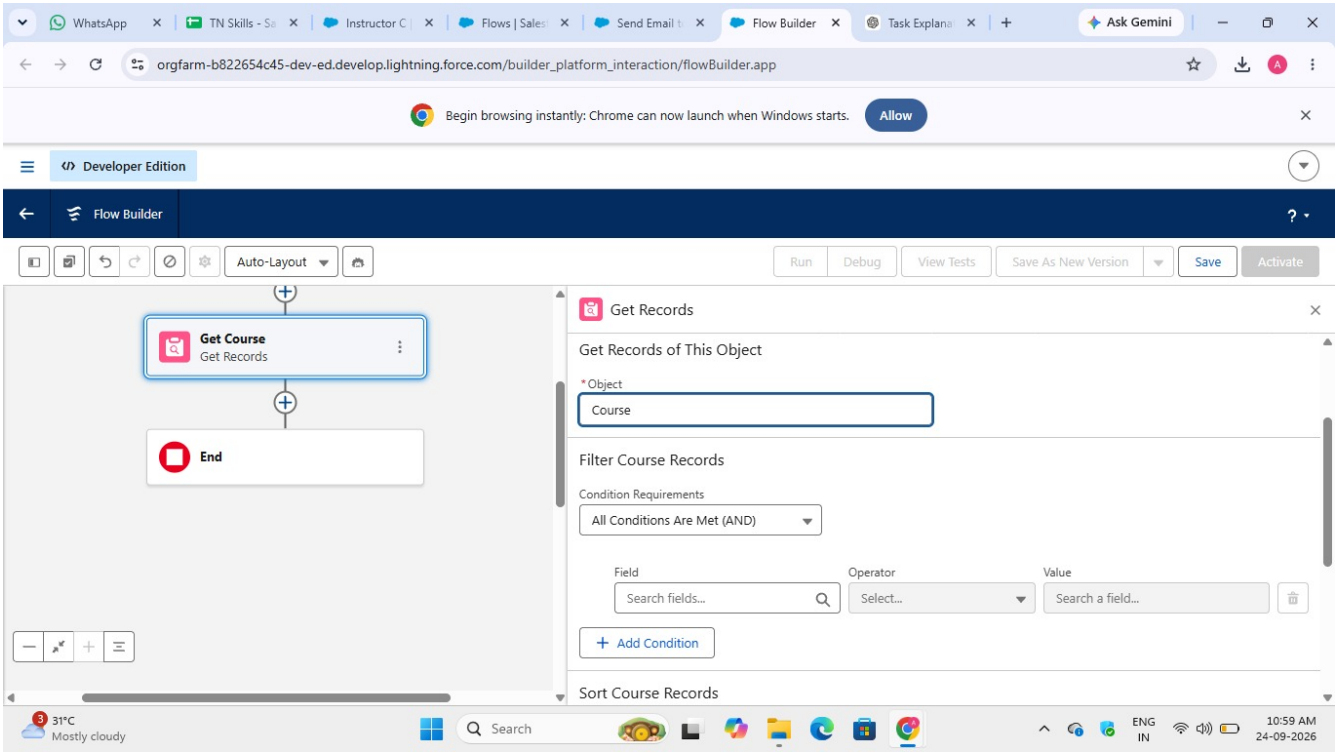
Salesforce Setup provides the Flows area for managing automation definitions.

Evidence 13. Record-triggered Flow start configuration



The Flow Builder start panel is configured for a record-triggered Flow; the object selection is not visible as completed in this capture.

Evidence 14. Get Course element

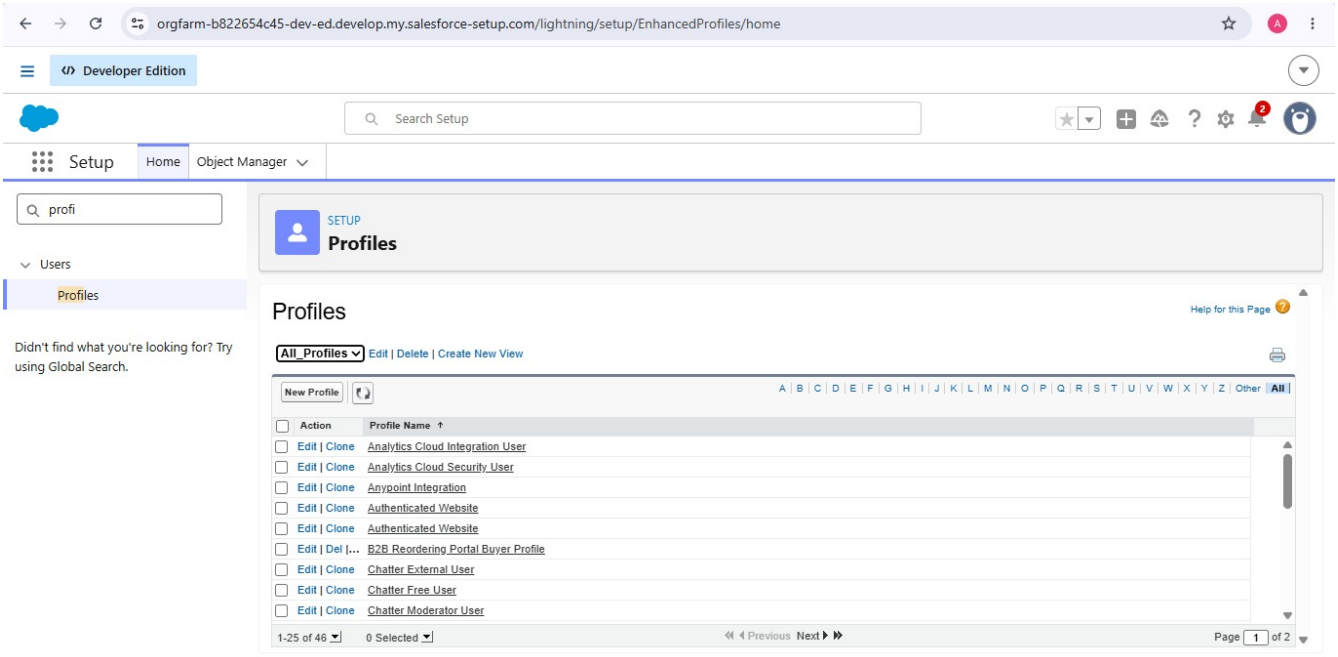


A Get Records element is configured to retrieve Course records. The displayed filter condition appears unfinished in this screenshot.

6.5 User Access and Security

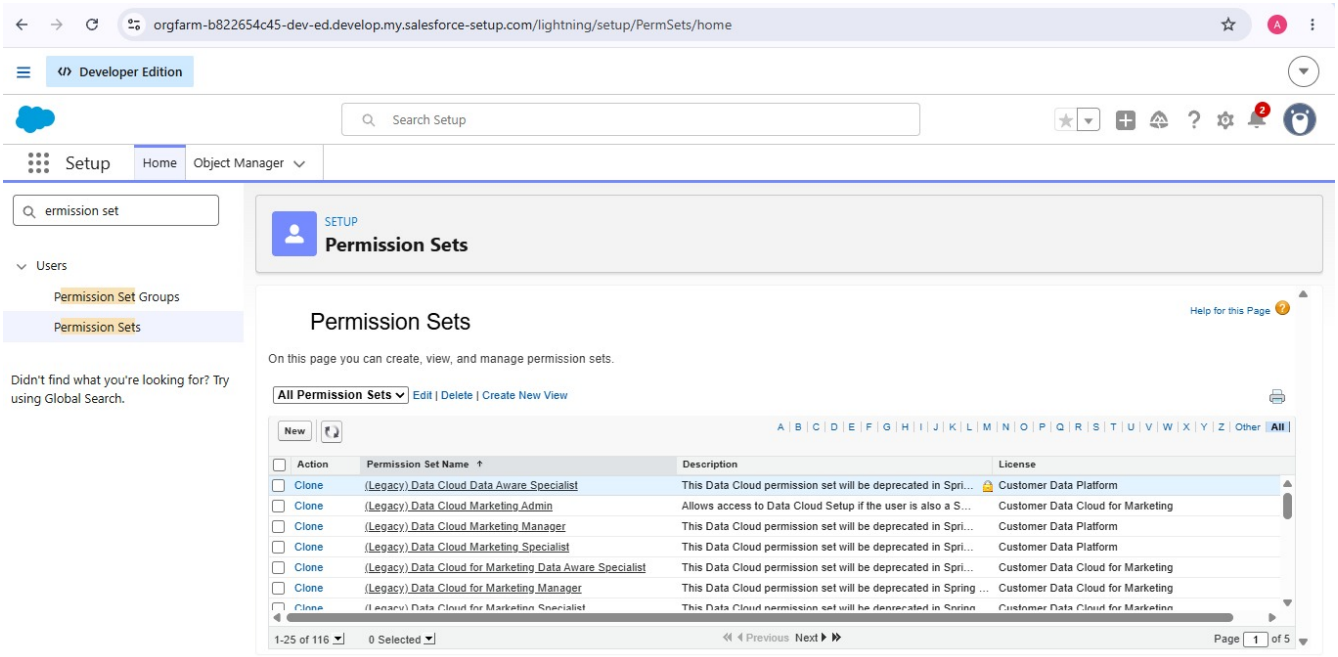
Profiles and Permission Sets provide the basis for granting access to project objects and functions. The design calls for broad configuration access for administrators, operational access for enrollment officers, and appropriately limited enrollment visibility for instructors. Organization-Wide Defaults and sharing rules should be set only after the institution confirms its access policy.

Evidence 15. Salesforce Profiles



The Setup page lists Salesforce profiles, where user-level permissions can be reviewed.

Evidence 16. Salesforce Permission Sets

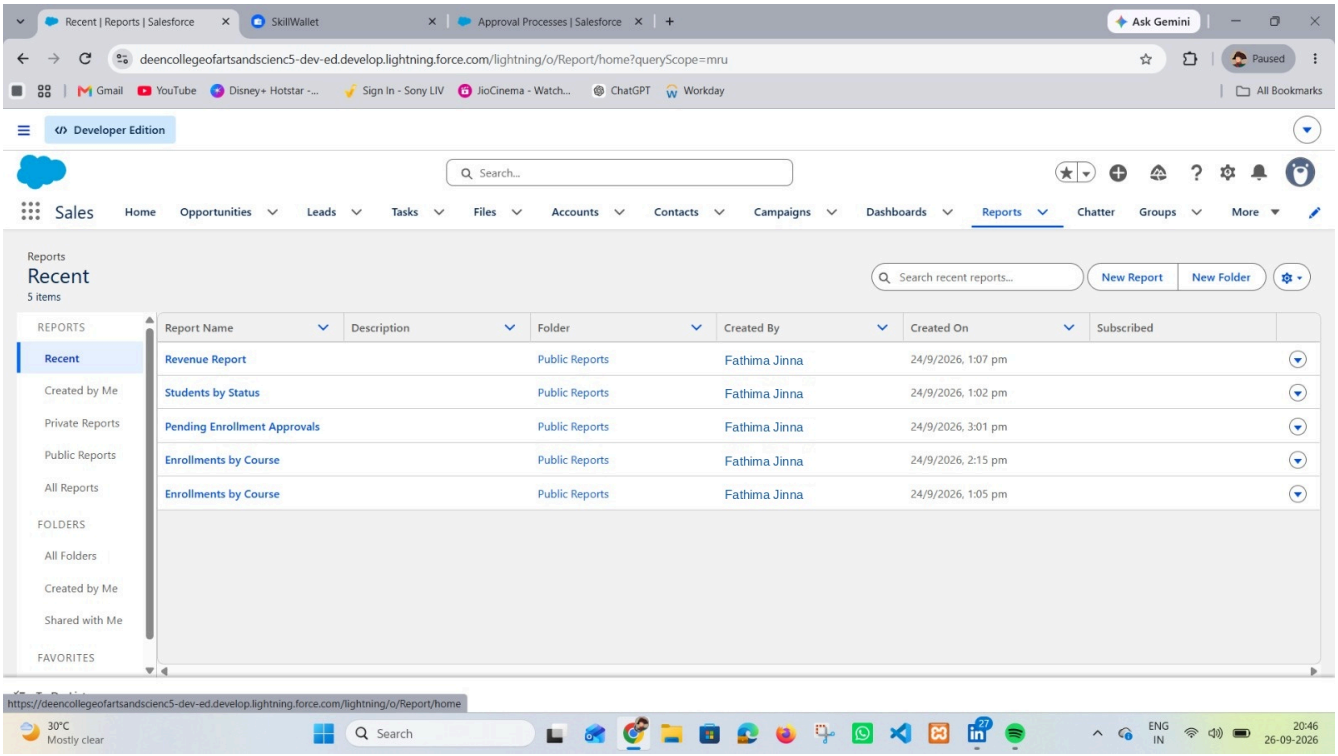


The Setup page lists Permission Sets for additional access assignments.

6.6 Reports and Dashboards

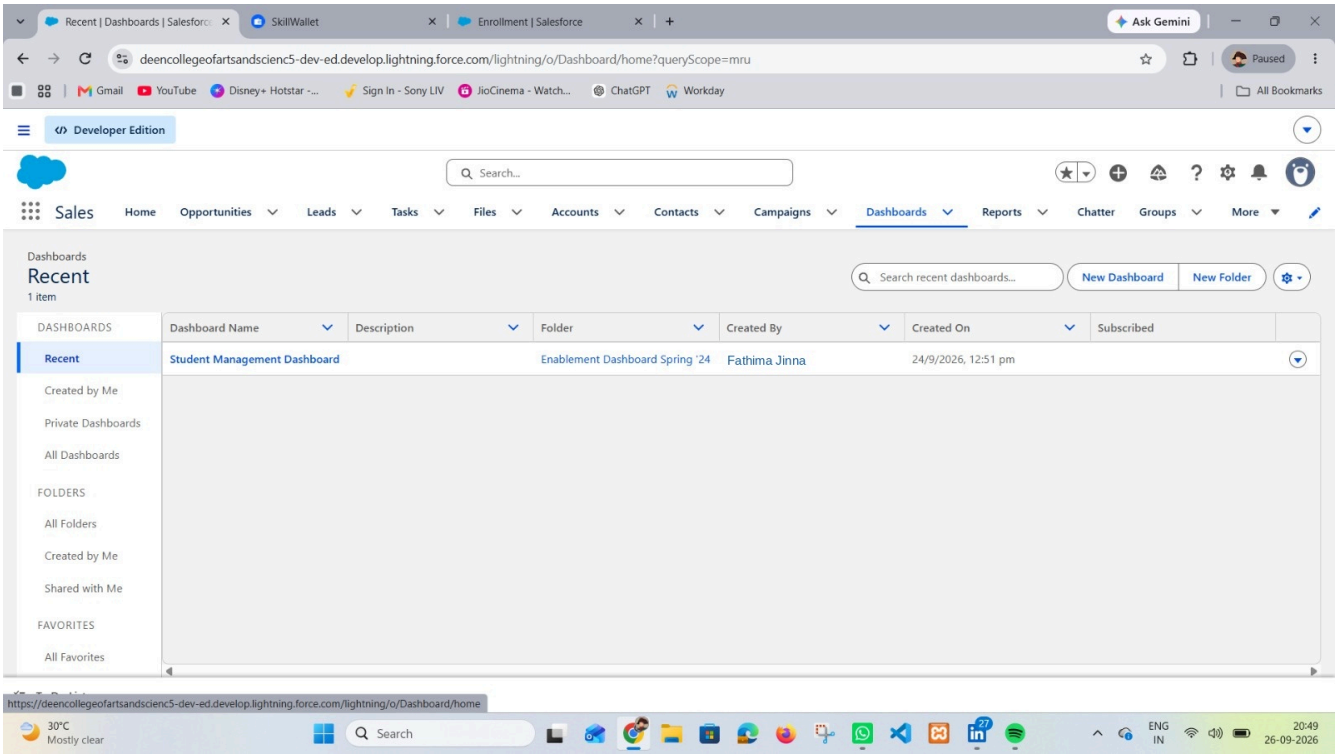
Reports provide filtered record-level views and summaries. The supplied project evidence includes a Students by Status report, and the report list shows additional enrollment and revenue report names. A dashboard can combine these report outputs to summarize student status and other operational measures. Dashboard results depend on the underlying report filters and available records.

Evidence 17. Salesforce report list



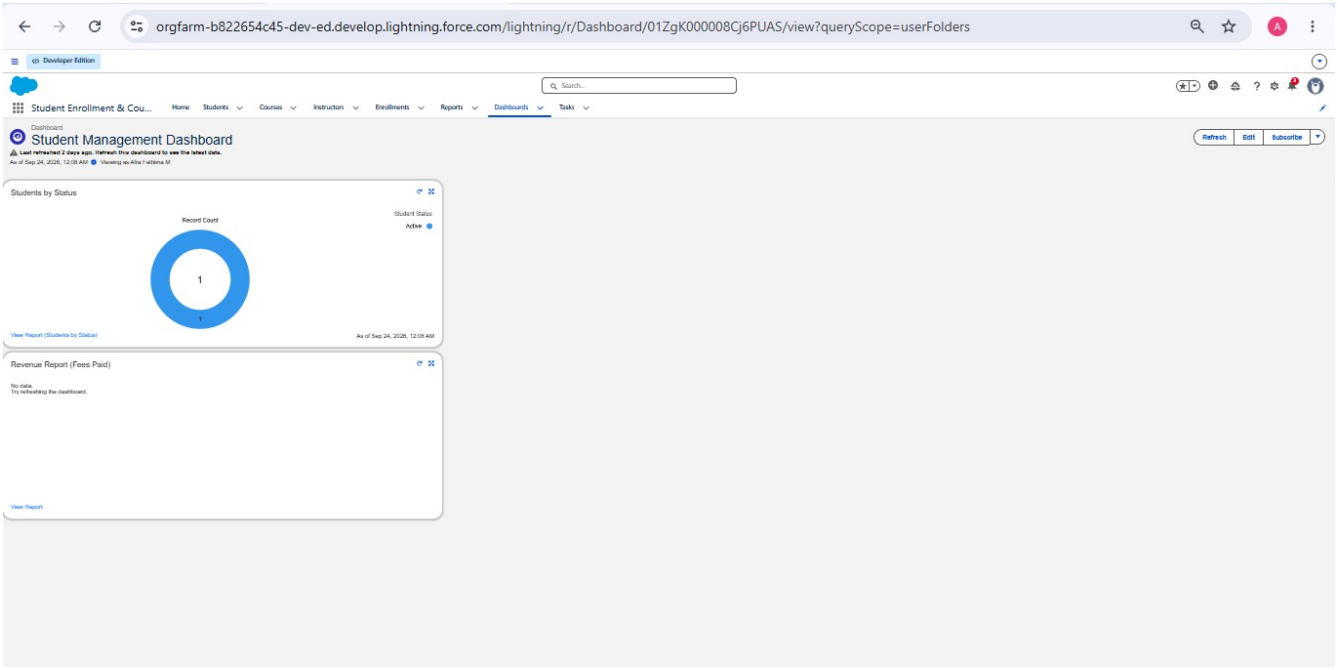
The report list contains Revenue Report, Students by Status, Pending Enrollment Approvals, and Enrollments by Course entries.

Evidence 18. Salesforce dashboard list



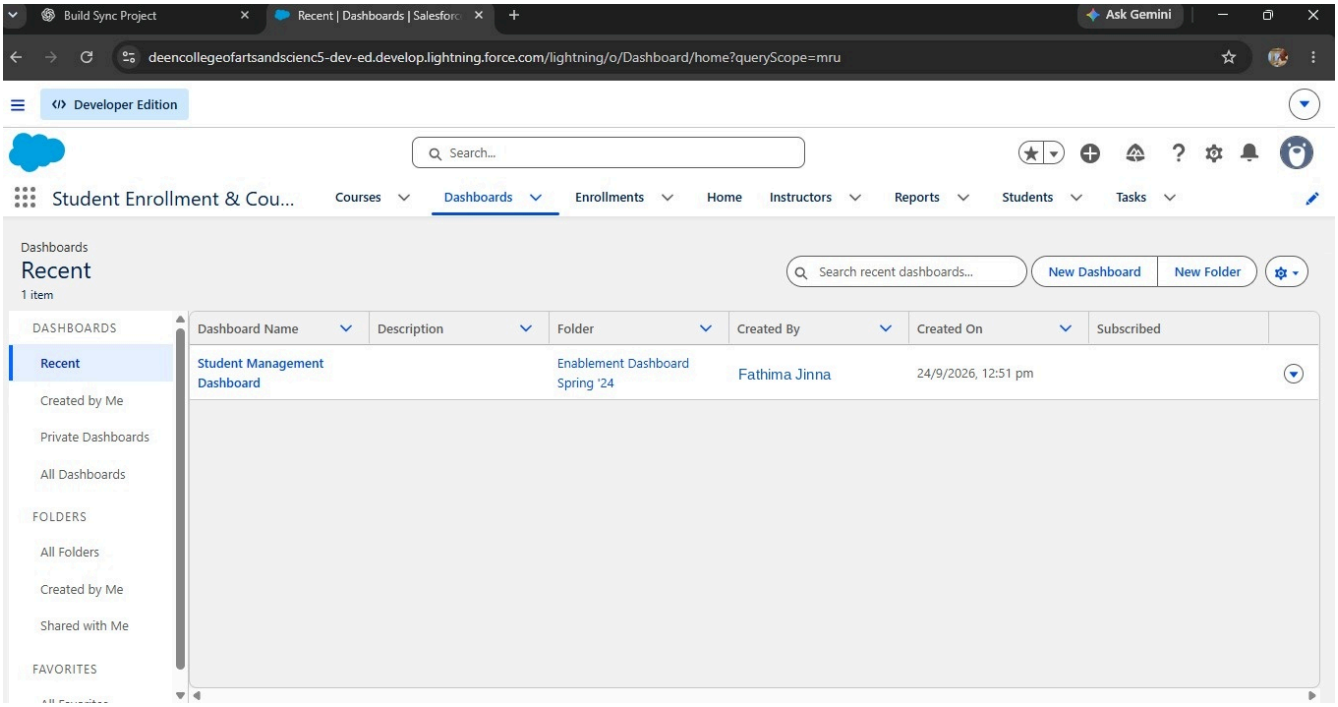
The dashboard list includes a Student Management Dashboard.

Evidence 19. Student Management Dashboard



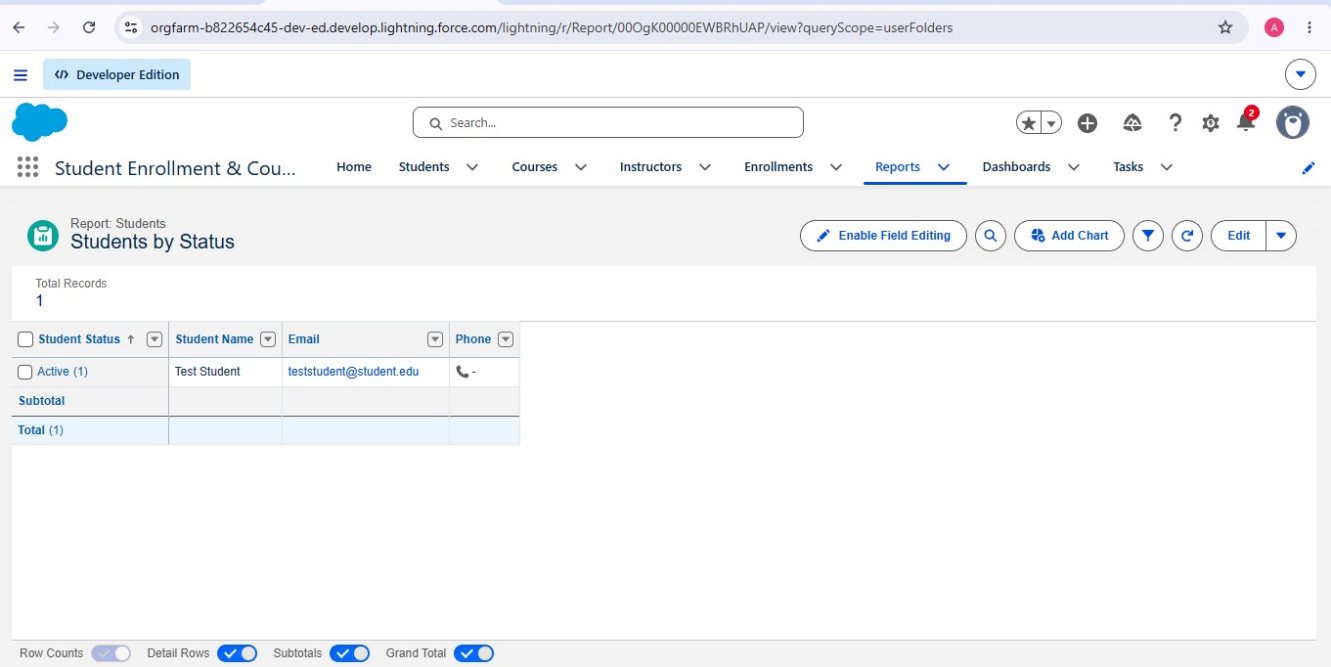
The dashboard displays a Students by Status chart and a Revenue Report component; the revenue component reports no data in this capture.

Evidence 20. Recent dashboards view



The Salesforce Dashboards page shows the Student Management Dashboard and its folder information.

Evidence 21. Students by Status report



The report shows one active student record with student name and email detail.

7. Functional & Performance Testing

7.1 Test Approach

Testing should cover record creation, relationships, validation, approval routing, Flow outcomes, reports, dashboards, and permissions. The evidence screenshots document visible configuration and sample report output; they do not, by themselves, establish that every intended scenario has passed.

Test ID	Test scenario	Expected result
TC-01	Create a Student, Course, and Instructor record.	Each record saves with valid required data.
TC-02	Create Enrollment and link Student, Course, Instructor.	Relationships are saved and visible on the record.
TC-03	Enter an invalid email or unpaid fee where prohibited.	Validation prevents saving or approval and explains the correction.
TC-04	Submit an Enrollment with Requested status.	The configured approval process routes the record to the assigned reviewer.
TC-05	Approve and reject representative requests.	The final status and configured notifications match each outcome.
TC-06	Create or update records that meet Flow criteria.	Only the intended automation runs and updates related information.
TC-07	Review report filters and dashboard components.	The displayed totals match the underlying records and filters.
TC-08	Access records using representative user roles.	Each role sees only the records and actions granted by policy.

7.2 Evidence-Based Observations

- The screenshots confirm the presence of the core Salesforce custom objects and several project fields.
- The report evidence shows one active student record in the captured status report.

- The dashboard screenshot shows a no-data state for the revenue component, indicating that additional suitable paid enrollment data may be needed to demonstrate that chart.
- The approval detail capture shows that an approval step was not yet defined at capture time.
- The Flow screenshots show a record-triggered Flow start and a Get Course element, while some configuration details appear incomplete.

These observations are recorded as shown by the supplied images. The team should execute and record the full test cases before presenting the system as fully validated.

8. Advantages and Limitations

8.1 Advantages

- Centralized student, course, instructor, and enrollment data improves record discovery.
- Connected records support enrollment-level reporting and related-list navigation.
- Validation, approvals, and Flows can reduce repeated manual work when fully configured.
- Reports and dashboards provide a shared view of status and operational activity.
- Salesforce configuration can be extended as requirements evolve.

8.2 Limitations

- The supplied screenshots show sample-scale data, including a single active student in one report.
- The approval detail screenshot does not show a defined approval step.
- Some Flow details appear unfinished in the supplied captures, so configured behavior should be verified.
- The revenue dashboard component shows no data in the captured state.
- The project evidence does not demonstrate a payment gateway, student self-service portal, or production deployment.
- Access policies, data privacy, backup, and deployment readiness require institution-specific review.

9. Conclusion

SECMS demonstrates how Salesforce can organize core enrollment information and provide a foundation for review, automation, and reporting. The Student, Course, Instructor, and Enrollment objects form the central data model. The screenshots document object configuration, fields, Flow setup, approval settings, and reporting views. Completing and testing the approval step, Flow criteria, sample data, and role-based access will strengthen the system and support a reliable demonstration.

10. Future Scope

- Complete and verify all approval steps, outcomes, and notifications.
- Finish Flow conditions and actions, then test them with representative records.
- Add student self-service for profile updates, course selection, and enrollment tracking.
- Integrate a secure payment service and reconcile payment status with enrollment records.
- Add capacity checks, waitlists, enrollment deadlines, and duplicate enrollment prevention.
- Improve analytics with term-based trends, course demand, and fee collection summaries.
- Strengthen data privacy, audit history, deployment controls, and user-specific sharing.

- Prepare a repeatable test plan and release checklist for future updates.

11. Appendix

11.1 Screenshot Index

The 21 supplied Salesforce screenshots are placed in the order of their related project topics: application and object setup; fields and relationships; approval and Flow configuration; security; then reports and dashboards. Captions describe what is visible in each image and distinguish completed evidence from configuration that appears incomplete.

11.2 References

- Salesforce Help, Salesforce Platform Documentation: https://help.salesforce.com/
- Salesforce Developers, Platform documentation: https://developer.salesforce.com/docs/
- Salesforce Trailhead, Salesforce learning modules: https://trailhead.salesforce.com/

11.3 Acknowledgement

We thank our project guide, department, institution, and classmates for their guidance and feedback during the development of the Student Enrollment & Course Management System.

11.4 Submission Details

Particular	Details
Project title	Student Enrollment & Course Management System (SECMS)
Presented by	Fathima Jinna, Nooris Raihan, Jumana Farveen, Shaafiya Hatheeja
Institution	Deen College of Arts and Science, Nidur
Academic year	2026
Guide / register numbers	Add the officially assigned details before final submission.